Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: I Semester - B. E CSE A

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Dr.M.Swarna Sudha , AP(SG)/CSE

Innovative Practice Description

Unit / Topic: Unit I / Guess an Integer number in a Range

Course Outcome: CO1

Topic Learning Outcome: TLO3

Activity Chosen: Role play

Date of Implementation: 13.10.2023

- **Justification:**

  Role play activities allow students to act out a scenario or take on the role of a person. Students can play out more complex scenarios by acting out these roles individually, in couples, or in groups. Students experience real-world scenarios through role plays. In role-playing exercises, participants take a role and behave in accordance with it.

  Guess an integer number in a range is a kind of game where the students are involved to play the game as per the rule. A user must guess a number between 0 and N in as little as 10 attempts in a simple guessing game known as a number guessing game. After ten tries, the player loses the game if he is unable to correctly guess the number. This method of active learning was chosen to assist students grasp the idea of using an algorithm to make an educated guess at an integer problem

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

  Faculty explained the concept of the problem. Three Pair students were asked to come in front of the classroom.

- The Role play of the guess number. First students generate the number between 1 to 10. The other students guess the number. if the both number is matched. The guesser wins the match. if not the generate has to give the clarification that the number is greater or small than the randomly generated number. The process will be repeated for 3 times Figure 1 explains the implementation of the game.
  - Sona S and Karthika A

- Arunadevi M and Sivapriya A
- Jeci Carmel A and Dhurka Devi S

The figure 2 shows the grouping and Students were made to play the game in the pair.

- Sona S was asked to fix a random number ‘x’ in the range of 1 to
- Karthika was asked to guess the number ‘y’. If the guessed number ‘y’ is greater than the random number ‘x’ then student 1 says “Too high”.
- If the guessed number ‘y’ is less than the random number ‘x’, then student 1 says “Too low”If the guessed number ‘y’ is equal to the random number ‘x’, then the student 1 says “You win ”To make the activity more interesting, number of guesses is fixed. If the student 2 fails to guess a number within the fixed number of guesses, the student 1 says “You lose”

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	**3**	**1**	**1**	**1**	**1**	**1**

(1 – Low   2 – Moderate   3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	It involves logical thinking and make the students able to think about problem solving approach	Students will be able to understand the problem in real manner than theoretical explanation	Students will be able to provide solution for the problem by writing python code through this activity	As the students are involved in the activity as team of two, the team work skill improved	Students communication skill will be improved as they are facing audience and communicating the activity	The problem solving skill earned through this activity Helps the students in solving various problems

Images / Screenshot of the practice:

Fig 1: Students involvement in Number Guessing Game

Figure 2: Group 1: Sona S and Karthika A Group2 : Arunadevi M and Sivapriya A Group 3: Jeci Carmel A and Dhurka Devi S

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - The students thought that this activity was very beneficial and gained valuable insight into the creation of an algorithm for predicting an integer number.
  - Students said it is quiet easy to understand when compare with other illustrative programs
  - Through this activity the students felt easy to write the program

- ***Benefit of the practice:***
  - Most of the students understood and wrote the step by step procedure of guessing an integer number individually.
  - Students interaction with their peer members and the class during the activity was good and effective

- ***Challenges faced in implementation:***
  - This activity involves only 6 members
  - Unable to make all the students to involve in this activity

References:

- https://serc.carleton.edu/introgeo/roleplaying/whatis.html
- https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf
- https://aminoapps.com/c/attack-on-titan/page/blog/different-types-of-roleplay-litteam/YoYg_nMTbux63E26M7z6xpRxkovvoBXPj